Vivek Preetham Mekala

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Professional Summary

Final-year Computer Science undergraduate specializing in Data Science & Analytics at IIIT Nagpur, with hands-on experience building and deploying production-grade AI/ML applications and Full-Stack AI systems. Demonstrated ability to architect end-to-end solutions spanning RAG pipelines, LLM integration, deep learning models, and scalable microservice backends — with projects live in production. Proficient in Python, JavaScript, and modern AI frameworks including LangChain, FastAPI, TensorFlow, and PyTorch, with a strong foundation in NLP and Deep Learning. Seeking to contribute to impactful AI/ML and Full-Stack AI roles at the intersection of intelligent systems and robust software engineering.

Education

Indian Institute of Information Technology, Nagpur — Nagpur, India
B.Tech in Computer Science & Engineering (Data Science and Analytics) — *Aug 2023 – June 2027*

Technical Skills

Languages: Python, C, C++, JavaScript
AI / ML & Data Science: TensorFlow, PyTorch, Keras, Scikit-learn, NLP, Deep Learning, Machine Learning, NumPy, Pandas, Matplotlib, Seaborn, OpenCV
Frameworks & APIs: Flask, FastAPI, Node.js, Express.js, Streamlit, REST APIs, LangChain
Databases: MySQL, MongoDB, FAISS
Tools & Platforms: Power BI, Tableau, Advanced Excel, Git, Groq, Vercel
Coursework: DSA, OS, OOP, DBMS, CN, ML, DL, NLP, Computer Vision, Big Data

Projects

DocuMind – RAG Document Intelligence Platform | *React, Node.js, FastAPI, FAISS, LangChain, Groq* GitHub

- Architected a full-stack **Retrieval-Augmented Generation (RAG)** platform enabling semantic PDF document Q&A, using a FastAPI microservice with LangChain, FAISS vector indexing, and Google Generative AI embeddings for context-aware retrieval.
- Built an end-to-end **PDF ingestion pipeline** using PyMuPDF and LangChain recursive chunking, converting document text into Google embeddings stored in FAISS for fast semantic search.
- Integrated **Groq-hosted LLM inference** for low-latency AI responses strictly grounded within uploaded document context, replacing local Ollama to enable cloud deployment.
- Designed a secure **MERN + Python hybrid backend** with JWT authentication, MongoDB Atlas chat persistence, and Multer-based file handling across a 3-service distributed architecture; deployed on **Vercel**.

ByteBrain – AI-Powered Personalized Learning Platform | *Flutter, Node.js, MongoDB, NLP, LLM* GitHub

- Engineered an end-to-end AI-powered learning platform adapting educational content based on learner preferences and proficiency levels, integrating an LLM-backed chatbot for intelligent content generation.
- Designed a modular application architecture combining retrieval strategies and candidate reranking, achieving up to **95% accuracy** with low-latency responses — directly analogous to RAG system design.
- Implemented analytics-driven personalization using NLP pipelines to process and serve contextually relevant learning material at scale.

Twitter Sentiment Analysis System | *Python, NLP, BiLSTM, TensorFlow*

- Developed a multi-class sentiment analysis solution leveraging BiLSTM with attention mechanism, improving classification accuracy from **92% to 95%** on a large-scale tweet corpus.
- Architected a comprehensive NLP preprocessing pipeline with exploratory data analysis to handle noisy social-media text, reducing neutral-class misclassification by approximately **15%**.
- Enhanced model interpretability by incorporating word-level attention, demonstrating ability to build and evaluate production-ready deep learning models.

Leadership & Activities

Senior Design Member, Abhivyakti & Design Lead, Gamers Night — Led visual branding, UI visuals, banners, and on-stage graphics for college-wide events.
Lead – Visual Arts, Design Club "Strokes" — Directed creative campaigns, illustrations, and design coordination for the club.